Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/17/2012
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Project Title:	Oxidative stress as a mechanism for altering acute and chronic functional changes in the CNS exposed to low dose, low dose rates, and mixed fields of protons and HZE nuclei		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
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No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: New end date is 6/30/2013 per A. Chu/ARC and NSSC information (Ed., 6/15/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Huang, Ting-Ting (VA Palo Alto Health Care System) Elmore, Eugene (University of California Irvine)		
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Task Description:	The space radiation environment poses unique hazards that include solar particle events and galactic cosmic rays. These types of radiations have been shown to cause acute and long-lasting chemical changes referred to as oxidative stress that can alter the normal physiology of the central nervous system (CNS). The chemical species produced can cause a wide range of effects in cells within the CNS that largely depend on the types of pathways they activate and the types of damage they produce. Proper functioning of the CNS depends on many factors, but is known to require subpopulations of neural stem and precursor cells that participate in the repair of CNS injury resulting from irradiation. Our proposal is focused on analyzing the effects of space radiation on these critical populations of multipotent neural stem and precursor cells. We have developed the technology to isolate and grow these cells in culture and will use a variety of techniques to analyze how cells exposed to space radiation change. We will monitor the time course of oxidative stress in these cells and how that depends on the dose, the dose rate and type of radiation applied. We will determine how radiation-induced oxidative stress impacts cellular proliferation, signaling between cells, cell fate and survival. Collectively our studies using both rodent and human neural stem and precursor cells will help NASA understand the acute and longer-lasting sequelae associated with the exposure of the CNS to the space radiation environment. These studies will facilitate efforts to more appropriately establish estimates of risk for radiation exposure to the CNS in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from these studies will be critical in enhancing our knowledge base of the risks associated with exposure of the brain to the radiation fields in space.
Task Progress:	We have initiated and completed several experiments proposed in our funded NASA grant. Our overarching goal is aimed at determining if/how low dose and low dose rate proton and HZE charged particle irradiation leads to oxidative stress in neural precursor cells, and whether oxidative stress constitutes a biochemical mechanism responsible for disrupting cellular physiology and promoting cognitive dysfunction. Our past data has demonstrated that the radiation response of multipotent neural precursor cells to x-rays, protons, and HZE nuclei involves an acute and persistent oxidative stress that can last for months. The importance of neural precursor cells to the repair and remodeling of the irradiated CNS is underscored by our data showing that the radiation-induced depletion of these cells is linked to the inhibition of neurogenesis and the onset of cognitive impairment. We believe oxidative injury is critical in this context, and plays a contributory if not causal role in determining neural precursor cell function, survival and fate. Our current progress extends these results considerably, as summarized below: 1. A comprehensive study has defined the radiobiological response of human neural stem cells to low LET radiation. Acharya, M., Lan, M., Kan, V., Patel, N., Giedzinski, E., Tseng, B.P. and Limoli, C.L. Consequences of ionizing radiation-induced damage in human neural stem cells. Free Radic. Biol. Med. 49:1846-1855 (2010). 2. A manuscript is under secondary review at Radiation Research that details the response of human and rodent neural stem cells to proton and gamma irradiation. Tseng, B.P., Lan, M.L., Tran, K., Giedzinski, E. and Limoli, C.L. Characterizing low dose and dose rate effects in rodent and human neural stem cells exposed to proton and gamma irradiation. Submitted Rad. Res. (2012). 3. We have published our results detailing the assessment of metabolic activity after low dose irradiation using non-invasive 2-photon spectroscopy. Krasieva, T.B., Giedzinski, E., Tran, K., Lan, M.L., Limoli, C.L. and Tromberg, B.J. Probing the impact of gamma-irradiation on the metabolic state of neural stem and precursor cells using dual-wavelength intrinsic signal two-photon excited fluorescence. Journal of Innovative Optical Health Sciences. 4, Britton Chance Special Issue 3:289-300 (2011). 4. We have published a review of the redox stress biology of irradiated neural stem cells. Globus, R.K., Caiozzo, V.J., Acharya, M., Fike, J.R. and Limoli, C.L. In: Oxidative Stress in Applied Basic Research and Clinical Practice: Cancer, Redox regulation of stem cell compartments: The convergence of radiation-induced normal tissue damage and oxidative stress. (Eds. D.R. Spitz, K.J. Dornfeld, D. Gius and K. Krishnan), Humana Press Inc. Totowa, NJ, pp.169-192 (2012). 5. Ongoing work with human neural stem cells is currently underway at BNL. We have participated at each of the NSRL runs in 2011 and plan to run experiments at NSRL-12A&B. 6. Significant work has been completed to ascertain the impact of altering the microenvironmental redox state using EC-SOD deficient neural stem cells. 7. Radioadaptive studies are ongoing and show that prior exposure alters radiation-induced oxidative stress. This stress has been found to in general, improve survival, enhance DSB repair, and elevate the intracellular antioxidant capacity of neural stem and precursor cells. Following up on our published work using hNSCs exposed to low dose gamma-irradiation (Acharya et. al. FRBM 49:1846, 2010 and Krasieva, et. al. J. Innov. Optical Health Sci. 3:289, 2011) we have undertaken significant work using low dose and low dose rate proton irradiation at Loma Linda. Low dose rate (LDR) protons were delivered at 20-25 cGy/hr whereas high dose rate (HDR) protons were delivered at 25-50 cGy/min. For gamma-irradiation, a 137Cs irradiator (J.L. Sheppard and Associates Mark I, CA, USA) was used at a dose rate of 2.2 Gy/min. We examined the effect of low dose and dose rate proton irradiation on human neural stem cells and found that changes in ROS/RNS levels were quite significant. At the 10, 25, and 50 cGy doses delivered at 20-25 cGy/hr, ROS/RNS levels increased significantly by 36, 56, and 37% respectively 6h after irradiation. A similar increase was not seen in cells receiving the high dose rate (25-50 cGy/min) where ROS/RNS levels were below controls at 6h post-irradiation and remained depressed until 36h. At 36h post-irradiation both dose rates resulted in significant increases in ROS/RNS levels. For LDR, 10, 25, 50, and 100 cGy doses increased ROS/RNS levels by 43, 57, 121, and 101% respectively, whereas for the same doses at HDR, ROS/RNS levels were increased further by 149, 134, 119, and 183% respectively. These increases were either sustained or elevated further by 48h, with levels at 100 to 300% above controls for both dose rates, with the largest increase (405%) found for the 10 cGy dose at HDR. Nitric oxide levels were also found to increase after most proton irradiations. At 6 and 12h post-irradiation, both dose rates gave qualitatively similar rises in nitric oxide levels, with peak levels observed at 24h. At this time, 10 cGy at LDR increased nitric oxide levels by 10%, while the remaining LDR doses elevated nitric oxide from 30 to 50%. At 24h, the HDR response revealed larger increases in nitric oxide, with levels elevated by 40 to 60% above control for all doses. Analyses of superoxide levels showed smaller fluctuations and were for the most part not significantly elevated (<10%)

	or decreased (<20%) compared to controls for both dose rates. Superoxide levels were however, significantly elevated at the higher doses used in these experiments. While a dose of 50 cGy at LDR elevated superoxide levels at 48h by 40%, a 100cGy dose at LDR was found to increase superoxide levels by 20 to 40% at 6, 12, 36, and 48h. A 100 cGy dose at HDR significantly increased superoxide levels by 20 to 40% at 6, 12, and 48h. We further investigated whether low dose and dose rate priming improved survival in the human neural stem cells. hNSCs were primed with both LDR and HDR 250 MeV protons. The hNSCs were given 25 cGy prior to 1 and 2 Gy gamma-irradiation challenge doses. Two trends were immediately noticeable. Both priming dose rates significantly improved survival following the 1 Gy challenge dose. Survival of 85% and 90% was found for the LDR and HDR respectively, compared to 68% for the non-primed cells. Despite this finding, neither priming dose was found to be beneficial for survival after a 2 Gy challenge dose. Survival of 34% and 20% was found for the LDR and HDR primed cells respectively, compared to 51% for the non-primed cells. When hNSCs were treated with both proton priming and challenge (HDR) doses similar trends were seen with the 1 and 2 Gy challenge doses. Lastly, we examined whether survival was improved after a priming dose of gamma-rays followed by 1 Gy challenge of HDR protons. Similar to prior results, we found that gamma-ray priming doses were sufficient to improve hNSC survival significantly (68%), compared to non-primed cells (46%). HNSCs cells subjected to low dose, high LET radiation exhibit elevated oxidative stress, similar to many of our past findings. HNSCs were also subjected to low dose iron ion irradiation using a range of energies from 300-1000 MeV/amu. Data analyzed at 1 and 7 days after irradiation show trends suggesting that at acute times, oxidative stress is more pronounced at higher particle energies while at chronic times oxidative stress is higher at lower particle energies. While preliminary, these findings suggest that acute effects may be more sensitive to the delta rays component of higher energy particles, while more chronic effects depend more heavily on particle LET. To more faithfully replicate the complexities of the space radiation environment we have undertaken a series of studies in which rodent neural stem and precursor cells were irradiated with proton and iron ions. Our strategy involved irradiating cells with either single or multiple doses of protons once a day over 4 days, then to subject them to a single HZE challenge dose of 600 MeV iron ions. The results of this study revealed that repeated irradiation caused a marked increase in oxidative stress as measured by each of our fluorogenic dyes, 3 days after the final dose. These data suggest that mixed ion fields in space pose risks for exacerbating radiation-induced oxidative stress. We have isolated a number of primary cultures of neural stem and precursor cells derived from EC-SOD deficient mice. Basal redox responses analyzed in unirradiated cells suggest that EC-SOD deficiency elicits a number of “expected” changes in redox state, owing to the higher baseline levels of superoxide in these cells. Such changes include increased superoxide levels (i.e. elevated Mitosox fluorescence), reduced nitric oxide levels (i.e. lower DAF fluorescence) and an overall increase in pro-oxidants (elevated CM/DCF fluorescence). Not unexpectedly, EC-SOD deficient neural stem and precursor cells subjected to heavy ion irradiation show elevated and persistent oxidative stress. At low dose exposures (= 1Gy) to 600 MeV iron ions EC-SOD deficient cells exhibit chronic oxidative stress when assayed 5 days post-exposure. At longer post-irradiation intervals, EC-SOD deficiency leads to lower levels of radiation-induced oxidative stress. Side-by-side comparison of WT and EC-SOD deficient cells assayed 1-week after low dose exposure to iron ions (600 MeV) shows lower levels of reactive species in the EC-SOD deficient cells. These findings suggest that elevated oxidative stress caused by SOD deficiency eventually leads to adaptation, and an attenuation of radiation-induced oxidative stress over longer post-irradiation intervals. In summary, we have obtained significant new data demonstrating that low dose irradiation can elicit oxidative stress in multipotent neural stem and precursor cells, derived from both rodents and humans. We have shown that several different redox sensitive fluorogenic dyes show significantly enhanced fluorescence following a range of low dose irradiation paradigms. Our data shows conclusively that prior exposure elicits adaptive responses that are capable of attenuating oxidative stress and survival. Collectively our data suggest that radiation-induced oxidative stress constitutes a biochemical mechanism of functional significance to neural stem cell biology.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Krasieva TB, Giedzinski E, Tran K, Lan ML, Limoli CL, Tromberg BJ. "Probing the impact of gamma-irradiation on the metabolic state of neural stem and precursor cells using dual-wavelength intrinsic signal two-photon excited fluorescence." Journal of Innovative Optical Health Sciences. Britton Chance Special Issue. 2011 Jul;4(3):289-300. http://dx.doi.org/10.1142/S1793545811001629 , Jul-2011
Books/Book Chapters	Globus RK, Caiozzo VJ, Acharya M, Fike JR, Limoli CL. "Redox regulation of stem cell compartments: The convergence of radiation-induced normal tissue damage and oxidative stress." in "Oxidative Stress in Cancer Biology and Therapy." Ed. D.R. Spitz, K.J. Dornfeld, K. Krishnan, D. Gius. New York : Springer, 2012. p.169-192., Jan-2012